

Jay Niketan Pathare

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EXPERIENCE

- Graduate Student Developer (CSE 611 Project)** | Team of 5 | HeinOnline, Buffalo *January 2026 – April 2026*
- Engineered a fault-tolerant OCR/LLM metadata extraction pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across three independent sources: CrossRef API, web scrapers, and OCR/LLM vision to extract article titles, authors, DOIs, page ranges, and ToC data from 47 law journals (28,167+ pages).
 - Built a fuzzy-matching combinator engine to merge asynchronous JSON outputs into a validated YAML structure, cutting manual data-entry workload by ~90% and raising extraction accuracy from 57.03% to 90.81% over five iterations (43 of 47 journals $\geq 80\%$, 8 at 100%; benchmarked against ground truth).
 - Parallelized the pipeline via a worker pool (cores – 1) that pulls the next journal from a shared queue on completion, and added an “aiscore” system to flag low-confidence fields for human review.
- Software Engineer** | Thesis Mumbai Tech (Startup), Mumbai, India *August 2024 – August 2025*
- Led delivery of platform modules end to end: gathered requirements in client meetings, prototyped to validate scope, decomposed the finalized work across a small team, and reviewed and merged their contributions into production.
 - Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records, and designed a real-time WebSocket + Redis pipeline feeding IoT baby-warmer sensor data into PostgreSQL with ~2s live monitoring.
 - Delivered a telemedicine consultation feature (WebRTC) connecting on-site nurses with remote doctors to review test results and procedures with patients, alongside secure PDF report generation.
 - Integrated a color-blindness diagnostic module (Ishihara plate testing, ~97–99% clinical sensitivity) with automated reporting and real-time result visualization for practitioners.
 - Architected relational schemas and SQL queries for efficient data retrieval, updates, and reporting.
- Cloud Engineer Intern** | Data Maven Pvt Ltd, Mumbai, India *November 2023 – May 2024*
- Designed and deployed scalable AWS infrastructure (EC2, RDS) with VPC networking and security groups, ensuring secure, low-latency communication for backend services.
 - Automated cloud resource provisioning, reducing setup time by ~40% and minimizing deployment-related issues.
- Data Engineer Intern** | Go Digital Technology Consulting, Mumbai, India *June 2023 – August 2023*
- Developed an ETL workflow using Python (Pandas, NumPy) and MySQL to process and clean large business datasets, generating reports highlighting key operational trends.

EDUCATION

- Master’s in Computer Science** *August 2025 – February 2027*
University at Buffalo, The State University of New York, Buffalo, NY **GPA: 3.85 / 4.0**
- Bachelor of Engineering, Information Technology** *August 2019 – May 2023*
Vivekanand Education Society’s Institute of Technology, Mumbai University, India

TECHNICAL SKILLS

- Languages & Databases:** Python, C, SQL, PostgreSQL, MySQL, HTML/CSS
- Frameworks & Libraries:** Django, Django REST Framework, Django Channels, ReactJS, PyTorch, scikit-learn
- AI/ML:** LLM Orchestration, Prompt Engineering, OCR (Tesseract), Machine Learning, Deep Learning, Model Evaluation, CNNs, GNNs (GraphSAGE, GAT), Transformers, Agentic AI
- Software Engineering:** OOP, REST APIs, Concurrency, Parallel Processing, Dynamic Programming, System Design, pytest, Unit & Metamorphic Testing, Load Testing, Agile, Parallel Processing, Real-Time Monitoring
- Tools & Cloud:** Linux, Docker, Docker Compose, Redis, AWS (EC2/RDS), Git, WebSockets, WebRTC, Locust, OAuth/Auth0, Gemini API, Groq API

ACADEMIC PROJECTS

- LLM Application Optimization Engine** | Individual Project *September 2026*
- Built a multi-model pipeline generating tailored resumes, cover letters, and independent fit assessments, using exact dynamic programming to select 3 of 6 projects across 60 valid allocations.
 - Reduced end-to-end runtime by ~73% (11 to ~3 minutes) through three concurrent LLM workers, account failover, and bounded retries, with deterministic checks rejecting unsupported claims.
 - Developed a 2,500+ test pytest suite with invariant and metamorphic tests for extraction, grounding, and optimization; verified generated PDFs by compiling LaTeX and re-extracting rendered text.
- Decoupled Learning Management System (LMS)** | Individual Project *July 2026*
- Architected a Django REST Framework and React LMS with Redis-backed Django Channels for live quizzes, leaderboards, and rate-limited chat, supporting 45+ concurrent users with sub-500ms latency.
 - Secured REST and WebSocket access with stateless JWT, implemented idempotent quiz grading, and load-tested with Locust and asyncio, reducing p95 REST latency by 26% and quiz-submission time from 2000ms to 1500ms.
- Fraud Detection using Graph Neural Networks** | Individual Project | University at Buffalo, NY *March 2026*
- Trained a GraphSAGE-based fraud detection model on the IEEE-CIS dataset (590k+ records), using Focal Loss to handle a severe 27:1 class imbalance and achieving a 0.9259 AUC-ROC (vs. 0.8584 MLP baseline, 0.7928 GAT).
 - Converted tabular transaction data into a graph network using shared card, device, and email identifiers, capping node groups to prevent memory explosion while capturing relational fraud rings.

CERTIFICATIONS, AWARDS & PUBLICATIONS

- AWS Cloud Practitioner** *July 2024*
- Certificate of Appreciation:** CSE Workshop “GitHub: Hands-On from Basics to Advanced,” UB *February 2026*
- Research Publication:** J. Pathare et al., “Music Genre Classification,” IJRAR, Apr. 2023 (97.68% accuracy). [Link](#).